

MONASH CLAYTON SCHOOL OF MEDICINE

2025 Year 1 Semester 2 Lecture Curriculum



MONASH University

Semester at a Glance

Yr 1 Semester 2 Lecture Timetable, 2025												
Theme	Week 1 (28-July)	Week 2 (4-Aug)	Week 3 (11-Aug)	Week 4 (18-Aug)	Week 5 (25-Aug)	Week 6 (1-Sept)	Week 7 (8-Sept)	Week 8 (15-Sept)	Week 9 (22-Sept)	Week 10 (6-Oct)	Week 11 (13-Oct)	Week 12 (20-Oct)
Personal & Professional Development	Health Enhancement Program	Health Enhancement Program	Health Enhancement Program									
	HEP - Positive Psychology 1 (JM)	HEP - Improving performance 1 (CHas)	HEP - Mental Health Enhancement (CHas)									
	HEP - Positive Psychology 2 (JM)	HEP - Improving performance 2 (CHas)										
Population, Society & Illness	Population Health POP - Introduction to Population Health (PB)			Population Health POP - Study Design (BD)		Population Health POP - Sampling and uncertainty (PR)		Population Health POP - Diagnosis and screening (EdV)	Population Health POP - Outbreak Investigation (SB)	Population Health POP - Life and death in the world (BD)		
Foundations of Medicine	Musculoskeletal/ Neuromuscular ANAT - Upper Limb 1 Introduction to the brachial plexus (DG)	Musculoskeletal/ Neuromuscular ANAT - Upper Limb 3 Compartments of the arm (DG)	Viral pathogens MICRO - Bacterial pathogenesis 1 (MH)	Musculoskeletal/ Neuromuscular ANAT - Overview Lower Limb - bones and articulation of the hip joint (DG)	Musculoskeletal/ Neuromuscular PHYS - Introduction to the autonomic nervous system (SC)	Musculoskeletal/ Neuromuscular PHYS - Pain (SC)	Viral pathogens MICRO - Viral pathogens 1 (MH)	Viral pathogens MICRO - Epidemiology & emerging infections (MH)	Musculoskeletal/ Neuromuscular ANAT - Overview of the back (DG)	Cardiovascular System - The Heart PHYS - Electrical activity of the heart 1 (JC)	Cardiovascular System - Blood Vessels PHYS - Arteries, arterioles, and the distribution of blood flow (JC)	Cardiovascular System - Blood Vessels PHYS - Coronary circulation in health and disease (BKH)
	ANAT - Upper Limb 2 Functional compartments and anatomy of the upper limb (DG)	PHYS - Intro to sensory systems (SC)	MICRO - Bacterial pathogenesis 2 (MH)	ANAT - Overview Lower Limb - the knee joint bones (DG)	PHYS - Autonomic nervous system - central systems (SC)	PHYS - Pain and temperature sensation (SC)	MICRO - Viral pathogens 2 (MH)	Musculoskeletal/ Neuromuscular PHARM - Pain, opioids & other analgesic agents (BB)	Cardiovascular System - The Heart ANAT - Embryology of the heart (MLaz)	PHYS - Electrical activity of the heart 2 (JC)	PHYS - Microcirculation, veins and venous return (JC)	PHYS - Vascular Endothelium (BKH)
	PHYS - Introduction to motor systems (SC)	PHYS - Neural integration and connections (SC)	MICRO - Bacterial pathogenesis 3 (MH)	ANAT - Overview Lower Limb - the ankle joint bones (DG)	PHYS - Discriminative somatic sensation (SC)	PHARM - Modulation of nerve conduction - local anaesthetics (BS)	MICRO - Infectious diseases of the skin and MSK system (MH)		ANAT - Overview of the thoracic cage and spaces (MLaz)	PHARM - Heart failure and inotropic agents (REW)	PHYS - Regulation of MAP and the cardiac cycle (JC)	PHARM - Atherosclerosis (REW)
	PHYS - Skeletal muscle (SC)	PHARM - Pharmacokinetics (MDeI)			PHARM - Autonomic nervous system - cholinergic and adrenergic Part 1 (BKH)	PHARM - Introduction to anti-inflammatory agents (BKH)			ANAT - Adult heart anatomy and innervation (MLaz)	PHARM - Antiarrhythmic agents (REW)	PHARM - Vasodilator agents (REW)	BIOCHEM - Lipid metabolism (NH)
					PHARM - Autonomic nervous system - cholinergic and adrenergic Part 2 (BKH)	PHARM - Steroidal anti-inflammatory drugs (BKH)			PHYS - Cardiovascular system overview and basic hemodynamics (JC)	PHARM - When cell die - cardiac biomarkers (REW)	PHARM - Pharmacology and physiology of stroke (BB)	PHARM - Lipid lowering drugs (REW)
											MICRO - Infection disease of the cardiovascular system (MH)	
							Human Development HD - Developmental Psychopathology (IL)	Human Development HD - Personality Development & Disorder (IL)	Human Development HD - Stress & Illness (IL)	Human Development HD - Prevention & Change (IL)	Human Development HD - Being Ill (IL)	Human Development HD - Pain (IL)
								HD - Social Cognition and moral development (IL)		HD - Becoming Ill (IL)		
Lecturer Key			Lecturer Key			Discipline Key						
BB	A/Prof Brad Broughton		NH	Dr Nathan Habila		ANAT	Antaomy					
BD	Prof Basia Diug		MLaz	Prof Michelle Lazarus		BIOCH	Biochemistry					
BKH	A/Prof Barb Kemp-Harper		PR	Ms Penny Robinson		HD	Human Development					
BS	Dr Ben Seyer		REW	Prof Rob Widdop		HEP	Health Enhancement Program					
CHas	Prof Craig Hassed		SC	Dr Simone Carron-Peters		MICRO	Microbiology					
DG	Dr David Gonsalvez		SB	Dr Sarah Borg		PHYS	Physiology					
EdV	Dr Elisha de Valle					PHARM	Pharmacology					
IL	Dr Isaac Lim					POP	Population Health					
JC	A/Prof Julia Choate											
JM	Dr Jo Mitchell											
MDeI	Dr Mark Del Borgo											
MH	A/Prof Meredith Hughes											

Week 1

Lecture 1

Theme: Personal and Professional Development

Session Outline: HEP - Positive psychology 1

Learning Outcomes
• Explore positive psychology theory and research • Apply ideas for building health and wellbeing

Lecture 2

Theme: Personal and Professional Development

Session Outline: HEP - Positive psychology 2

Learning Outcomes
• Explore positive psychology theory and research • Apply ideas for building health and wellbeing

Lecture 3

Theme: Population, Society & Illness

Session Outline: POP - Introduction to population health

Learning Outcomes
• Define epidemiology and describe how it is used in public health • Explain the concepts of “sufficient” and “necessary” in causation • Describe the different criteria for establishing causation including temporal relationship, plausibility, consistency, strength, dose-response relationship, reversibility and study design • Explain the concept of confounding and describe how to control for confounding in order to help establish causation

Lecture 4

Theme: Foundations of Medicine

Session Outline: ANAT - Upper limb 1: Introduction to the brachial plexus

Learning Outcomes
Part 1 • Relate the structure and development of the brachial plexus to upper limb function
Part 2 • Describe the concept of a dermatome and identify the dermatomes of the upper limb • Describe the concept of a myotomes and identify the spinal cord segments responsible for movements of the upper limb • Describe how spinal cord segments are distributed within the brachial plexus and its major terminal branches
Part 3 • Link the shape of articular surfaces with the function of muscles acting on joints to explain the movements of the pectoral girdle and shoulder joint

Lecture 5

Theme: Foundations of Medicine

Session Outline: ANAT - Upper limb 2: Functional compartments and anatomy of the upper limb

Learning Outcomes
Part 1 • Integrate knowledge of shoulder function with possible findings on clinical examination
Part 2 • Describe the organisation of the deep fascia of the arm and its clinical relevance (e.g. compartment syndrome) • Identify muscles that cross multiple joints and describe why this makes them susceptible to injury (e.g. long head of biceps brachii)

Lecture 6

Theme: Foundations of Medicine

Session Outline: PHYS - Introduction to motor systems

Learning Outcomes
• Understand what makes up the motor system and explain the importance of sensory information for motor control • Describe the connectivity and role of the upper and lower motor neuron systems • Explain the role and connectivity between different brain regions and pathways involved in initiating, refining and controlling movement • Describe motor units and the how force generation in muscle is controlled via motor units

Lecture 7

Theme: Foundations of Medicine

Session Outline: PHYS - Skeletal muscle

Learning Outcomes
• Describe the structure, components and organization of the muscle • Explain the molecular mechanisms of skeletal muscle contraction, cross bridge cycling, the sliding filament mechanism and excitation-contraction coupling • Explain the mechanics of muscle contraction • Define the different types of muscle contractions - isotonic, isometric, concentric and eccentric

Week 2

Lecture 1

Theme: Personal and Professional Development

Session Outline: HEP - Improving performance 1

Learning Outcomes
• Understand stress, motivation, attention and performance • Understand attentional challenges – Technology – Distractor influence – Multitasking • Understand selective attention and inattention blindness • Understand mindful practice and training attention

Lecture 2

Theme: Personal and Professional Development

Session Outline: HEP - Improving performance 2

Learning Outcomes
1. Decision making • Understand System 1 and 2 thinking • Understand utility theory • Understand cognitive bias • Understand mindsets
2. Creativity
3. Bloom, Miller and Bigg • Understand Bloom's levels of thinking • Understand Miller's levels of performance • Understand Bigg's model of learning
4. Highly effective people • Understand team learning • Understand time management • Understand Covey's 7 Habits of Highly Effective People

Lecture 3

Theme: Foundations of Medicine

Session Outline: ANAT - Upper limb 3: Compartments of the arm

Learning Outcomes
Part 1 • Link the shape of articular surfaces with the function of muscles acting on joints to explain the movements of the elbow joint • Integrate knowledge of elbow function with possible findings on clinical examination (e.g. supracondylar fracture) • Identify relations of neurovascular structures around the distal arm and elbow and their clinical relevance
Part 2 • Classify the intrinsic muscles of the hand according to their functions, innervation and clinical testing
Part 3 • Describe the course and distribution blood vessels in the forearm and hand, and relate the course of these vessels to clinical applications • Describe the course and distribution of the median, ulnar and radial nerves in the forearm, wrist and hand, and relate the course of nerves to potential to sites of injury

Lecture 4

Theme: Foundations of Medicine

Session Outline: PHYS - Introduction to sensory systems

Learning Outcomes
• Explain what makes up the sensory system, and how are they activated to create perception of the five senses • Describe how sensory receptors transduce a physical stimulus into a biology response by changing the membrane potential • Explain how sensory receptors code for stimulus properties (primary sensory coding)

Lecture 5

Theme: Foundations of Medicine

Session Outline: PHYS - Neural integration and connections

Learning Outcomes
• Describe the functional need for integration of information within the nervous system • Explain the processes and pathways of integration • Describe the locations of integration within the nervous system • Apply how brain lesions and maladaptive circuitry and neural connections contribute to functional deficits

Lecture 6

Theme: Foundations of Medicine

Session Outline: PHARM - Pharmacokinetics

Learning Outcomes
• Describe how drugs are absorbed, distributed, metabolized and excreted from the body • Discuss how a knowledge of pharmacokinetics of a drug is important when considering the route of administration, the dosing schedule, and ways of achieving therapeutic levels whilst avoiding toxicity • Explain the terms pharmacokinetics, therapeutic index, bioavailability, volume of distribution, half life, clearance, plasma protein binding, biotransformation, first order kinetics, zero order kinetics, induction of liver enzymes

Week 3

Lecture 1

Theme: Personal and Professional Development

Session Outline: HEP - Mental health enhancement

Learning Outcomes
• Understand issues and challenges related to doctor and medical student health, and aspects of medical culture that can contribute to poor mental health • Define burnout, work engagement and resilience • Apply effective strategies to enhance workplace wellbeing, resilience, mindfulness, behavioural activation and emotional intelligence • Have strategies in place for being able to intervene with troubled peers

Lecture 2

Theme: Foundations of Medicine

Session Outline: MICRO - Bacterial pathogenesis 1

Learning Outcomes
• Describe the different mechanisms by which bacteria invade the human body and interact with host cells • Appreciate how pathogens evade and overcome the host immune system • Relate how bacteria cause the host damage that leads to the clinical signs of disease

Lecture 3

Theme: Foundations of Medicine

Session Outline: MICRO - Bacterial pathogenesis 2

Learning Outcomes

- Describe the different mechanisms by which bacteria invade the human body and interact with host cells
- Appreciate how pathogens evade and overcome the host immune system:
 - Avoid being killed by phagocytosis
 - Living inside phagocytic cells
 - Mechanism of invasion of non-phagocytic cells
 - Bacterial movement within and between cells
- Relate how bacteria cause the host damage that leads to the clinical signs of disease

Lecture 4

Theme: Foundations of Medicine

Session Outline: MICRO - Bacterial pathogenesis 3

Learning Outcomes

- Describe the different mechanisms by which bacteria invade the human body and interact with host cells
- Appreciate how pathogens evade and overcome the host immune system
- Relate how bacteria cause the host damage that leads to the clinical signs of disease: i.e.
 - septic shock
 - toxins that activate T cells
 - toxins that activate the host cell surface
 - toxins that modify intracellular targets

Week 4

Lecture 1

Theme: Population, Society & Illness

Session Outline: POP - Study design

Learning Outcomes
• Identify, describe and differentiate between descriptive and analytical study designs • Evaluate the advantages and disadvantages of different study designs • Describe the study design that best suits your study objectives and can test your hypothesis

Lecture 2

Theme: Foundations of Medicine

Session Outline: ANAT - Overview of the lower limb: Bones and articulations of the hip joint

Learning Outcomes
Part 1: • Describe the osteology and arthrology of the hip joint
Part 2: • Describe the vascular supply of the gluteal region and compartments of the thigh
Part 3: • Describe the function and innervation of muscles in the gluteal region, and in the compartments of the thigh • Integrate knowledge of the hip joint and thigh with possible findings on clinical examination

Lecture 3

Theme: Foundations of Medicine

Session Outline: ANAT - Overview of the lower limb: The knee joint bones

Learning Outcomes
Part 1 • Describe the osteology and the arthrology of the knee joint • Integrate knowledge of the knee and leg anatomy with possible findings on clinical examination
Part 2 • Integrate knowledge of the knee and leg anatomy with possible findings on clinical examination

Lecture 4

Theme: Foundations of Medicine

Session Outline: ANAT - Overview of the lower limb: The ankle joint bones

Learning Outcomes
Part 1 • Describe the osteology and arthrology of the tibiofibular joints
Part 2 • Describe the function and innervation of muscles in the compartments of the leg • Describe the vascular supply of the leg • Integrate knowledge of leg anatomy with possible findings on clinical examination
Part 3 • Integrate knowledge of ankle and foot anatomy with possible findings on clinical examination

Week 5

Lecture 1

Theme: Foundations of Medicine

Session Outline: PHYS - Introduction to the autonomic nervous system

Learning Outcomes
• Describe the components and the overall role of the autonomic nervous system • Compare and contrast between somatic and autonomic nervous systems • Differentiate between the structures and functions of the sympathetic and parasympathetic divisions in the autonomic nervous system

Lecture 2

Theme: Foundations of Medicine

Session Outline: PHYS - Autonomic nervous system: Central systems

Learning Outcomes
• Describe the different levels of central control on the autonomic nervous system • Understand the role of the different central nuclei (such as hypothalamus, brain stem, amygdala etc.) involved in autonomic regulation • List some examples of autonomic reflexes that are modulated by central control centres

Lecture 3

Theme: Foundations of Medicine

Session Outline: PHYS - Discriminative somatic sensation

Learning Outcomes
• Identify types of somatic sensations • Identify and categorize the skin layers and its receptors • Compare and contrast the four main types of mechanoreceptors • Describe proprioception and proprioceptors

Lectures 4 and 5

Theme: Foundations of Medicine

Session Outline: PHARM - Pharmacology of the autonomic nervous system

Learning Outcomes
• Describe the steps in the neurotransmission pathways in the ANS • Identify sites where drugs can act to modulate this process & the mechanisms by which they do this • Explain the advantages/disadvantages of these sites as therapeutic targets • Predict and explain the effects of drugs that act at these sites

Week 6

Lecture 1

Theme: Population, Society & Illness

Session Outline: POP – Sampling and uncertainty

Learning Outcomes
• Describe the concept of random sampling from a population • Explain the concept of sampling variability • Describe the principles of how statistical theory is used to quantify the uncertainty in the result of a single study • Explain the difference between population parameters and sample statistics • Explain the following terms: standard error of the mean, confidence interval for a mean, confidence interval for the difference between two means • Interpret confidence intervals in published research papers

Lecture 2

Theme: Foundations of Medicine

Session Outline: PHYS - Pain

Learning Outcomes

- Understand that nociceptive signals from the periphery are processed in the spinal dorsal horn, before transmission to the thalamus and cortical regions
- Recognise that descending pathways from the higher brain centers can modulate nociceptive transmission in the dorsal horn
- Understand the differences between acute and chronic pain and that that pain can be generated pathologically without nociceptor activation.
- Have a basic understanding of the mechanisms that drive chronic neuropathic and inflammatory pain syndromes
- Understand how visceral pain is generated and why it can be referred to the surface of the body
- Know the different classes of analgesic drugs and where they act in the nociceptive system

Lecture 3

Theme: Foundations of Medicine

Session Outline: PHYS – Pain and temperature sensation

Learning Outcomes

- Understand somatic sensation: pain and temperature sensations
- Know that these sensations are detected by un-encapsulated receptors and thin diameter primary afferent fibres
- Recognise the pathways to the cerebral cortex for pain, temperature and touch sensations
- Know the properties of thermoreceptors
- Know the properties and types of nociceptors
- Describe how nociceptors are activated and sensitized

Lecture 4

Theme: Foundations of Medicine

Session Outline: PHARM - Modulation of nerve conduction: Local anaesthetics

Learning Outcomes
• Describe the mechanism of action of the local anaesthetics • Describe the different properties of the various local anaesthetics • Describe the methods of administration of local anaesthetics • Describe the unwanted effects of local anaesthetics

Lecture 5

Theme: Foundations of Medicine

Session Outline: PHARM - Introduction to anti-inflammatory agents

Learning Outcomes
• Describe the main symptoms of an inflammatory response • List the main mediators responsible for these symptoms • List the different classes of medication used to relieve pain and inflammation • Describe the mode of action of non-opioid analgesics (e.g. simple analgesics and NSAIDs) • Describe the major side effects associated with the use of non-opioid analgesics

Lecture 6

Theme: Foundations of Medicine

Session Outline: PHARM - Steroidal anti-inflammatory drugs

Learning Outcomes
• List the different classes of medication used to relieve pain and inflammation • Explain the physiological role of endogenous steroids • Discuss the anti-inflammatory actions of the glucocorticosteroids • List the common side-effects associated with the long-term use of the glucocorticosteroids

Week 7

Lecture 1

Theme: Foundations of Medicine

Session Outline: MICRO - Viral pathogens 1

Learning Outcomes
• Recall viral structure and obligate intracellular lifecycle • Recall transmission of viral infections • Describe the major outcomes of viral infection • Describe how viruses are classified by their characteristics and the Baltimore classification system • Define and order the key steps in the viral life cycle • Give examples of the pathogenesis of some important DNA and RNA human viruses

Lecture 2

Theme: Foundations of Medicine

Session Outline: MICRO - Viral pathogens 2

Learning Outcomes

- Recall viral structure and obligate intracellular lifecycle
- Recall transmission of viral infections
- Describe the major outcomes of viral infection
- Describe how viruses are classified by their characteristics and the Baltimore classification system
- Define and order the key steps in the viral life cycle
- Give examples of the pathogenesis of some important DNA and RNA human viruses

Lecture 3

Theme: Foundations of Medicine

Session Outline: MICRO - Infectious disease of the skin and MSK system

Learning Outcomes

- Relate the three main mechanisms of how skin and soft tissue infections occur
 - Breach, Systemic infection, Toxin mediated
- Describe bone and joint infections
- Appreciate that the majority of skin and MSK infections are caused by viruses and bacteria, but there can also be fungal and parasitic causes
- Recognise the significance of the bacteria, *Streptococcus pyogenes* and *Staphylococcus aureus* as important common causes of many skin and MSK infections

Lecture 4

Theme: Foundations of Medicine

Session Outline: HD - Developmental psychopathology

Learning Outcomes
• Summarise three broad principles for defining atypical or abnormal behaviour and the approach taken in the DSM-5 to diagnosing psychological disorders • Explain how abnormal cognition and behaviour may be an outcome of development rather than disease • Explain the diathesis-stress model of psychopathology • Explain the influence of biopsychosocial factors on developmental psychopathology across the lifespan

Week 8

Lecture 1

Theme: Population, Society & Illness

Session Outline: POP - Diagnosis and screening

Learning Outcomes
• Identify the role of population screening tests as tools of public health • Explain the concepts of false positive and false negative results • In relation to a test result, explain the meaning of sensitivity, specificity, positive and negative predictive values • Identify and describe the strengths and weakness of screening with a focus on over-diagnosis and treatment • Recognise and cite evidence to support the rationale behind various targeted prevention programs

Lecture 2

Theme: Foundations of Medicine

Session Outline: MICRO - Epidemiology and emerging infections

Learning Outcomes

- Relate the importance of infectious disease epidemiology
- Define key terminology such as:
 - Latent period, incubation period, infectious period, clinical disease
 - Morbidity, mortality
 - Endemic, epidemic, pandemic
 - R_0 , transmission terminology
 - Herd immunity
- Appreciate the steps used in an outbreak investigation to break the chains of transmission
- Recognise the importance of emerging and re-emerging infections and the conditions that may cause them to arise

Lecture 3

Theme: Foundations of Medicine

Session Outline: PHARM - Pain, opioids and other analgesic agents

Learning Outcomes
• Describe the clinical uses of opioid analgesics and related agents • Outline the mechanism of action of opioid analgesics and related agents • Explain the major side effects associated with opioid analgesics and related agents • Describe the analgesic ladder

Lecture 4

Theme: Foundations of Medicine

Session Outline: HD - Personality development and disorder

Learning Outcomes
• Outline basic concepts and theories related to self and personality • Identify and describe The Big Five personality dimensions • Describe the role of culture in the development of self-conceptions • Identify the key characteristics of personality disorders • Discuss the biopsychosocial influences on personality and personality disorders

Lecture 5

Theme: Foundations of Medicine

Session Outline: HD - Social cognition and moral development

Learning Outcomes
• Describe social cognition and outline how theory of mind unfolds during infancy and childhood and the significance for later development • Describe how nature and nurture contribute to the development of a theory of mind and perspective taking • Explain how well social cognition and perspective taking holds up in adulthood and later life • Define and illustrate the three components of morality • Compare and contrast the main messages about morality according to psychoanalytic, cognitive developmental, social learning and evolutionary theorists

Week 9

Lecture 1

Theme: Population, Society & Illness

Session Outline: POP - Outbreak investigation

Learning Outcomes
• Define commonly used terms associated with outbreaks and epidemics • Be aware of the 10 steps of outbreak investigation (NOTE: Many steps often occur concurrently!) • Interpret epidemic curves (cases per day, and total cases)

Lecture 2

Theme: Foundations of Medicine

Session Outline: ANAT - Overview of the back

Learning Outcomes
Part 1 • Describe the general structure of the vertebral column and contrast regional variations in its bony components • Discuss the structure of a typical vertebra, and describe and contrast the special features of cervical, thoracic and lumbar vertebrae
Part 2 • Describe the curvatures of the vertebral column and discuss their abnormalities • Identify and describe the joints and ligaments of the vertebral column • Describe the movements of the vertebral column
Part 3 • Identify and locate the principal extrinsic and intrinsic muscles of the back • Describe the movement and function of the back muscles

Lecture 3

Theme: Foundations of Medicine

Session Outline: ANAT - Embryology of the heart

Learning Outcomes

- Define the anatomical differences between the adult heart and the foetal heart
- Explain anatomical structures facilitating heart development
- Compare and contrast typical and atypical foetal heart development identifying the impacts on post-natal heart anatomic topology
- Compare and contrast the foetal versus post-natal circulation. Include in this discussion the role of pressure changes within the heart, and the impact of other visceral structures (i.e. lungs and liver) on cardiac circulation
- Outline the embryological precursors for adult heart anatomy, including the embryological and post-natal terminology associated with these structures

Lecture 4

Theme: Foundations of Medicine

Session Outline: ANAT - Overview of the thoracic cage and spaces

Learning Outcomes

- Describe the functions of the thoracic wall and how the anatomy supports this function
- Identify thoracic surface anatomy which can help you in diagnosis and treatment
- Explain thoracic cage anatomy, and identify key relationships
- Discuss thoracic anatomical structures integral to the thorax's function in each of the following:
 - Protection
 - Space
 - Coordinating Movement & Decreasing Friction

Lecture 5

Theme: Foundations of Medicine

Session Outline: ANAT - Adult heart anatomy and innervation

Learning Outcomes

- Illustrate the relationship between physical exam surface anatomy and the anatomical topography of the heart and major vessels
- Compare and contrast the anatomical location of heart valves to their auscultation location in physical exam
- Explain the layers surrounding the heart, and the relationship between these layers and clinical presentations (i.e. pericarditis and cardiac tamponade)
- Describe the perfusion pathway to (and away) from the cardiac tissue (i.e. coronary vessels), and explain how this pathway relates to the cardiac cycle
- Illustrate the innervation of the heart, and the functional impacts of this innervation on the heart

Lecture 6

Theme: Foundations of Medicine

Session Outline: PHYS – Cardiovascular system overview and basic hemodynamics

Learning Outcomes

- Explain the overall design of the human cardiovascular system and its functions
- Explain the basic anatomy of the heart and its arrangement into 4 chambers
- Appreciate that blood flows in series through the systemic and pulmonary circulations
- Appreciate that blood flows in parallel through many different vascular beds' circulations
- Identify which regions of the cardiovascular system contain blood high in oxygen content and which regions contain blood low in oxygen content

Lecture 7

Theme: Foundations of Medicine

Session Outline: HD - Stress and illness

Learning Outcomes
• Define the concept of stress • Apply contemporary models of stress to clinical examples • Explain how the impact of events on individuals' stress levels can be increased or decreased by: ◦ personality traits ◦ cognitive factors ◦ social support ◦ emotions • Define coping • Categorise coping strategies using models of coping

Week 10

Lecture 1

Theme: Population, Society & Illness

Session Outline: POP - Life and death in the world

Learning Outcomes

- Describe patterns of health and disease on a world-wide scale including expected trends over the course of the next century
- Source and interpret bio-statistical information relevant to health and disease in populations across a range of contexts
- Describe the main risk factors associated with premature death to understand the basis of health promotion programs
- Describe the relationship between exposure, causation and risk factors for disease and morbidity in populations

Lecture 2

Theme: Foundations of Medicine

Session Outline: PHYS - Electrical activity of the heart 1

Learning Outcomes

- Describe the conducting system of the heart and how electrical activity spreads from the sinoatrial node to the rest of the heart
- Appreciate that action potentials in different regions of the heart have different shapes and the reasons for this
- Describe the main ionic movements during a ventricular action potential
- Explain how sinoatrial node pacemaker cells spontaneously generate action potentials

Lecture 3

Theme: Foundations of Medicine

Session Outline: PHYS - Electrical activity of the heart 2

Learning Outcomes
• Describe the three phases of the ECG, including how they relate to excitation of the heart and the cardiac action potentials • Describe some common cardiac arrhythmias • List pacemaker cells/regions within the heart • Explain why the sinoatrial node acts as the normal pacemaker of the heart • Explain how sinoatrial node pacemaking (i.e. heart rate) is regulated by the autonomic nervous system

Lecture 4

Theme: Foundations of Medicine

Session Outline: PHARM - Heart failure and inotropic agents

Learning Outcomes
• Know the signs & causes of heart failure • Understand the pathophysiology of heart failure • List major drug classes used in the treatment of heart failure • Describe mechanisms of action of drugs used in treatment of heart failure & adverse effects • Have knowledge of ABD + inotropes e.g. digoxin

Lecture 5

Theme: Foundations of Medicine

Session Outline: PHARM - Anti-arrhythmic agents

Learning Outcomes
• Describe cardiac rhythm and types of arrhythmias (basic understanding) • Describe major drug classes used in the treatment of cardiac arrhythmias (Class I - IV) • Explain mechanisms of action and adverse effects of antiarrhythmic drugs and when used

Lecture 6

Theme: Foundations of Medicine

Session Outline: PHARM - When cells die: Cardiac biomarkers

Learning Outcomes
• Explain acute coronary syndrome (ACS) and acute myocardial infarction (AMI) diagnoses • Remember main biomarkers & sensitivities used in diagnosis of AMI; in particular, troponin • Understand use of biomarkers (e.g. CK, troponin)/ECG for prediction and management of ACS/ AMI • Understand the use of BNP and C-reactive protein (CRP) as biomarkers for heart failure (HF) and MI-related CV events, respectively • Compare/contrast AMI & chronic heart failure with respect to cardiac remodeling (revisit when discussing HF)

Lecture 7

Theme: Foundations of Medicine

Session Outline: HD - Prevention and change

Learning Outcomes
• Apply the Health Action Process Approach to basic clinical examples • Define attitudes and describe the dimensions along which they vary • Identify factors in persuasive communication • Appreciate the benefits of using imagery in health communication • Apply the elaboration likelihood model, at an introductory level

Lecture 8

Theme: Foundations of Medicine

Session Outline: HD - Becoming ill

Learning Outcomes
• Contrast symptoms and signs • Explain the process of symptom perception and apply it to clinical examples • Understand the impact of variables on symptom perception • Describe the Common-Sense Model of self-regulation in symptom and risk perception • Recognise the five components of an Illness Representation • Define attribution and describe the dimensions along which attribution can vary • Identify common errors in attribution

Week 11

Lecture 1

Theme: Foundations of Medicine

Session Outline: PHYS - Arteries, arterioles and the distribution of blood flow

Learning Outcomes

- Explain why arterial blood pressure is pulsatile and define the terms diastolic, systolic, pulse and mean arterial pressure
- Understand the importance of the elastic nature and low resistance of the large arteries
- Discuss how arteriole diameter and blood flow can be controlled by local (intrinsic) factors (metabolic and myogenic control)
- Explain how sympathetic nerves (noradrenaline via α receptors) and hormonal adrenaline (via vascular smooth muscle alpha and beta-2 receptors, with differing effects) alter arteriolar diameter and the distribution of blood flow

Lecture 2

Theme: Foundations of Medicine

Session Outline: PHYS - Microcirculation, veins and venous return

Learning Outcomes

- Describe the transport pathways for capillary exchange
- Explain the factors that influence filtration and absorption at the arteriolar and venular ends of a capillary (Starling forces - hydrostatic and osmotic pressures)
- Discuss how altered capillary dynamics lead to extracellular oedema
- Explain the role of the lymphatic system for returning filtered fluids to the circulation

Lecture 3

Theme: Foundations of Medicine

Session Outline: PHYS - Regulation of MAP and the cardiac cycle

Learning Outcomes
• Explain the relationship between mean arterial pressure (MAP), cardiac output (CO) and total peripheral resistance to flow (TPR) • Discuss how baroreceptors are involved in the short-term neuronal control of blood pressure (i.e. the arterial baroreflex) • Explain how standing up suddenly or fainting can lead to a drop in MAP and HR, and how the body acts (via afferent sensory and efferent autonomic neuronal pathways) rapidly to return MAP back to normal • Summarise the events occurring in the different phases of the cardiac cycle (e.g. pressure and volume changes, opening and closure of valves, timing of blood filling and blood ejection) • Relate the phases of the cardiac cycle to the electrocardiogram (ECG)

Lecture 4

Theme: Foundations of Medicine

Session Outline: PHARM - Vasodilator agents

Learning Outcomes
• Recognise the signs and causes of angina • List major drug classes used in the treatment of angina • Describe mechanisms of action of drugs used in treatment of angina & adverse effects: (focus on nitro-vasodilators)

Lecture 5

Theme: Foundations of Medicine

Session Outline: PHARM - Pharmacology and physiology of stroke

Learning Outcomes
• Describe the incidence, risk factors and acute symptoms of stroke • Describe the cellular and vascular pathophysiology of stroke, including the "ischaemic cascade" • Identify the types of drugs used in the treatment of stroke, and explain their mechanisms and sites of action

Lecture 6

Theme: Foundations of Medicine

Session Outline: MICRO - Infectious diseases of the cardiovascular system

Learning Outcomes
• Describe the major infectious diseases of the cardiovascular system focusing on: • Infectious endocarditis, myocarditis, pericarditis • Provide a microbiological and clinical framework for these infections: • Give examples of the common microbial agents of disease • Relate the pre-disposing conditions that can lead to these types of infections • Indicate some of the signs, symptoms and tests used to diagnose these CVS infections • Describe fever of unknown origin

Lecture 7

Theme: Foundations of Medicine

Session Outline: HD - Being ill

Learning Outcomes
• Recognise delay behaviour and the motivations behind delay behaviours • Appreciate the broad impact of being ill on the individual and family and social networks • Comprehend the sick role and the limitations of this concept • Comprehend the role of carers and consider both the advantages and disadvantages on patients and carers

Week 12

Lecture 1

Theme: Foundations of Medicine

Session Outline: PHYS - Coronary circulation in health and disease

Learning Outcomes
• Understand the gross anatomy of the coronary circulation • Appreciate the high reliance of the heart on an adequate oxygen supply and the role of the coronary circulation in providing this • Describe the unique features of the regulation of the coronary circulation • Describe the outcomes of atherosclerosis involving the major coronary arteries • State the major determinants and risk factors for coronary atherosclerosis • Explain various treatment modalities

Lecture 2

Theme: Foundations of Medicine

Session Outline: PHYS - Vascular endothelium

Learning Outcomes
• Explain the importance of the vascular endothelium in modulating vascular smooth muscle tone • Identify endothelium-derived vasoactive factors and their mechanisms of action • Explain the consequences of damage to the vascular endothelium

Lecture 3

Theme: Foundations of Medicine

Session Outline: PHARM - Atherosclerosis

Learning Outcomes
• Remember key processes involved in development of atherosclerosis/ plaque formation • Understand mechanisms that contribute to plaque rupture (NOT venous thrombus formation) • Explain potential CVD complications arising from plaque rupture • Explain treatment for acute life-threatening events

Lecture 4

Theme: Foundations of Medicine

Session Outline: BIOCH - Lipid metabolism

Learning Outcomes
Part 1 • Outline the relevance of essential fatty acids • Describe in outline the role of lipoproteins • Explain the importance of ketone bodies
Part 2 • Describe what lipoproteins are and explain in outline their role in cholesterol metabolism
Part 3 • Explain the importance of ketone bodies

Lecture 5

Theme: Foundations of Medicine

Session Outline: PHARM - Lipid lowering drugs

Learning Outcomes
• Describe the processes involved in lipoprotein transport • Describe the processes involved in development of atherosclerosis and subsequent risks; e.g. familial hypercholesterolaemia • Describe the non-pharmacological treatment of elevated lipid levels • Describe the basic mechanism of action of lipid-lowering drugs: statins, fibrates, cholesterol absorption inhibitors • Appreciate newer drug targets: e.g. PCSK9

Lecture 6

Theme: Foundations of Medicine

Session Outline: HD - Pain

Learning Outcomes
• Be familiar with the definition and classifications of pain and to understand the sensory, emotional and cognitive aspects of pain • Know the prevalence of pain • Be aware of the major impacts of persistent pain on mood, behaviour, function and quality of life as well as recognise the important role of psychological factors in mediating these impacts • Apply gate control theory to basic clinical situations • Select an appropriate methods of pain assessment • Understand the major approaches to the management of persistent pain problems